

Actuator Optimal Design

Lecture 4 : Actuator Design Process(上)

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Actuator Design Process

- Define Requirements
- Define Torque-Mass-Inertia(T-M-I)
- Select Reducer Type
- Design Reducer
- Design Structural Components

Define Requirements

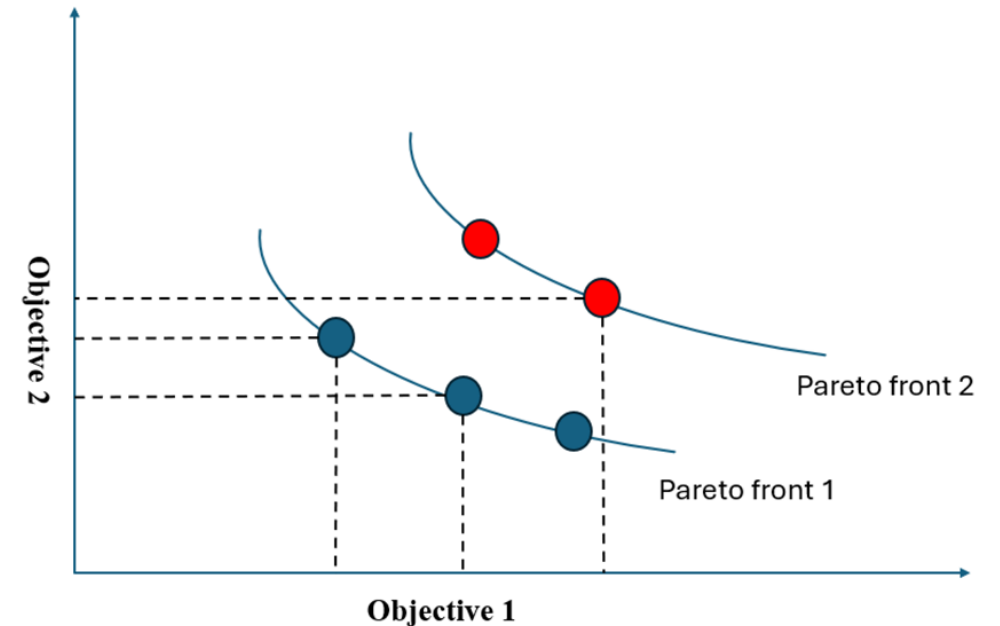
Define Requirements

- Requirements(or, Conditions) **MUST BE PRE-DEFINED!!!**
Choose the optimal point on the current Pareto front first,
BEFORE trying to push the Pareto front forward.
- Optimal requirement selection method is **INDEPENDENT** of actuator optimal design.

- **Example**

To design a mountain bike:

- Select MTB structure first,
- Then optimize materials.



Define Requirements

- Actuator Primary Requirements

Mass / Rated Torque / Inertia / Peak(Saturation) Torque → 4DoF

→ **Fundamentally** related to each other.

→ To improve one, the other one must be worsened.

- Actuator Secondary Requirements

Backlash / Volume / Stiffness / Velocity(Power)...

→ No Fundamental relation, **Practical** limit.

HOW?

Define Requirements

- Actuator Primary Requirements

Mass / Rated Torque / Inertia / Peak(Saturation) Torque → 4DoF

→ **Fundamentally** related to each other.

→ To improve one, the other one must be worsened.

$$\frac{K_m^2 \tau_{sat}^2}{M I^2} \approx c$$

K_m : Motor constant($K_t/\sqrt{R_s}$)

τ_{sat} : Torque limit due to core saturation

M : Mass of copper

I : Joint inertia

- For off-the-shelf frameless motor(when you select it, Not design it),

K_m and τ_{sat} are coupled with **1 DoF**.

→ For off-the-shelf frameless motor, just consider **Torque / Mass / Inertia, 3DoF**.

Define Requirements

Example procedures for a **QDD**-based **anthropomorphic** manipulator:

1. Define kinematic parameters.
 2. Set Rated / Peak Payload.
 3. Define Torque / Mass / Inertia of *Wrist* (3DoF) actuators.
 4. Define Mass of *Forearm*.
 5. Define Torque / Mass / Inertia of *Elbow* (1DoF) actuator.
 6. Define Mass of *Upper arm*.
 7. Define Torque / Mass / Inertia of *Shoulder* (3DoF) actuators.
- Total mass is too heavy?
 - return to Step x and repeat iteratively.
 - Failure to meet goals?
 - Lower goals(payload, total mass) or push Pareto front(better motor, material, gear)

Define Requirements

Example procedures for a **QDD**-based **anthropomorphic** manipulator:

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HOW?

- Total mass is too heavy?
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Define Torque-Mass-Inertia(T-M-I)

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- **Set Rated / Peak Torque**

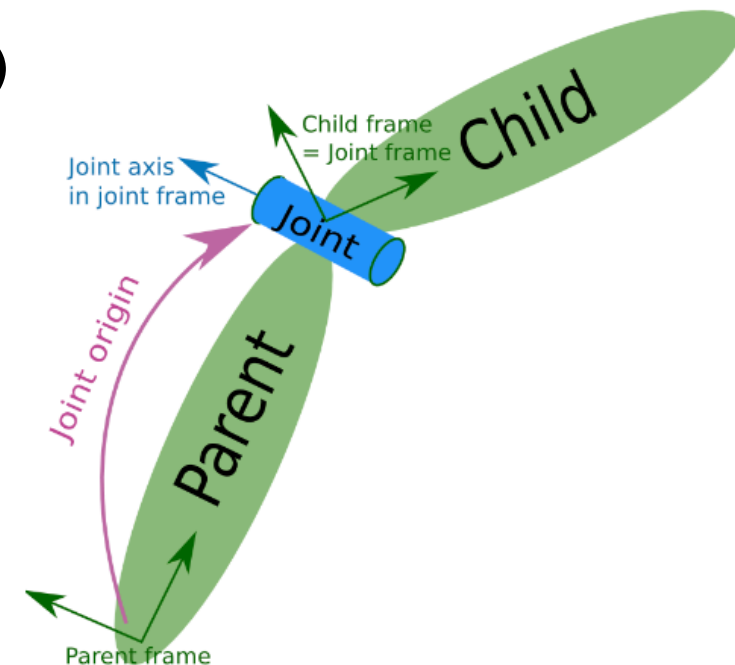
Use Jacobian with payload and child link mass ($\tau = J^T F$).

This is Easy, but which **configuration** should be considered?

→ For fixed-base high precision manipulators, **WORST CONFIGURATION**.

→ For mobile-base, anthropomorphic manipulators, **FAVORABLE CONFIGURATIONS**.

Now, you set Rated / Peak **Torque** of your **actuator(NOT MOTOR!!!)**



+ Very roughly, Peak Torque is **300–400%** of Rated Torque.

Define Torque-Mass-Inertia(T-M-I)

- **Set Mass**

Example conditions for a **QDD**-based **static, anthropomorphic** manipulator:

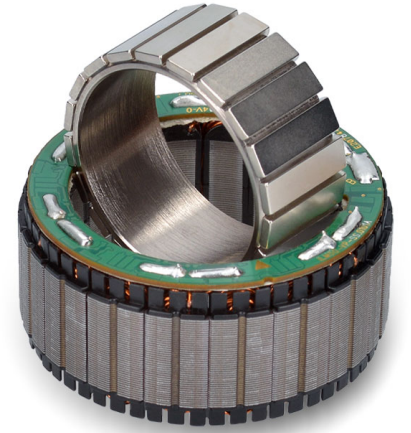
(Assumptions)

→ When you use off-the-shelf frameless motors(No custom design).

→ Focused on static performance, NOT dynamic performance.

1. Lighter motors are better.
2. Maximize reduction ratio (joint torque / motor torque) within reduction ratio limit.
Reduction ratio limit is set by manufacturing quality & gear design optimization.
3. Check joint **velocity** with given battery voltage.

+ Mass of reducer greatly depends on Peak joint torque, not motor torque.



Define Torque-Mass-Inertia(T-M-I)

- **Inertia & Reduction Ratio(Static)**

Inertia is often neglected when dynamic performance **IS NOT** critical.

Excessive inertia (=high reduction ratio) reduces back-drivability.

→ Reduction ratio is typically limited to **~20:1**.

With optimal reducer(+gear tooth profile) design,

→ Back-drivability can be maintained even at **~100:1[1]**.

Define Torque-Mass-Inertia(T-M-I)

- **Inertia & Reduction Ratio(Dynamic)**

When dynamic performance is critical,

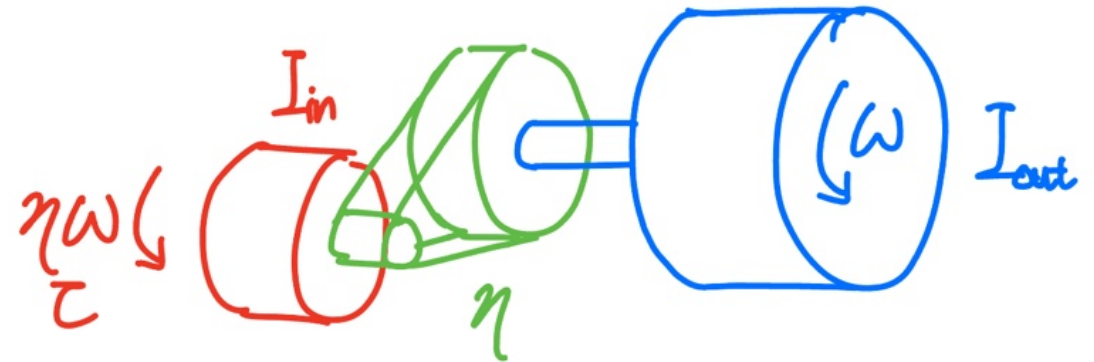
→ Joint acceleration should be maximized.

For a 1-DoF system,

→ Maximum acceleration is achieved when **Reflected joint inertia = link inertia.**

$$\begin{aligned}(\tau - I_{in}\eta\dot{\omega})\eta &= I_{out}\dot{\omega} \\ \eta\tau &= (I_{out} + \eta^2 I_{in})\dot{\omega} \\ \dot{\omega} &= \frac{\eta}{I_{out} + \eta^2 I_{in}}\tau\end{aligned}$$

$$\begin{aligned}\frac{d}{d\eta} \left(\frac{\eta}{I_{out} + \eta^2 I_{in}} \right) &= (I_{out} + \eta^2 I_{in})^{-1} - 2\eta^2 (I_{out} + \eta^2 I_{in})^{-2} I_{in} = 0 \\ I_{out} &= \eta^2 I_{in}\end{aligned}$$



Select Reducer Type

Select Reducer Type

Robot Actuator Reducer Types:

- Planetary Drive(single-stage, 2-stage, 3K-compound...)
- Cycloidal Drive
- Harmonic Drive

- Reducer Primary Requirements

Torque density / Efficiency

→ Fundamental Trade-off exists!

- Reducer Secondary Requirements

Backlash

→ No Fundamental relation, **Practical** limit.

Select Reducer Type

- What is 'Back-drivability'?

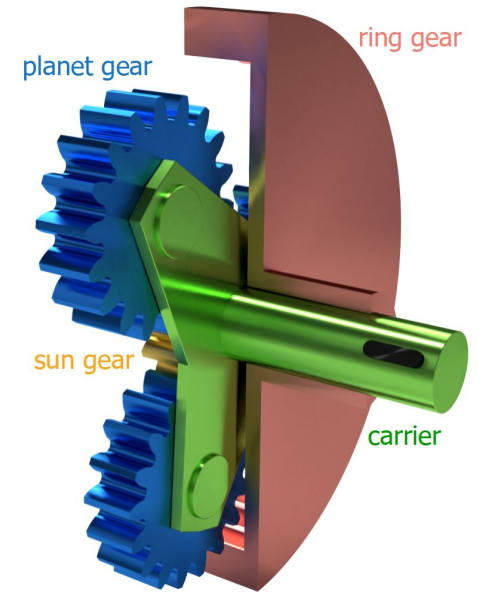
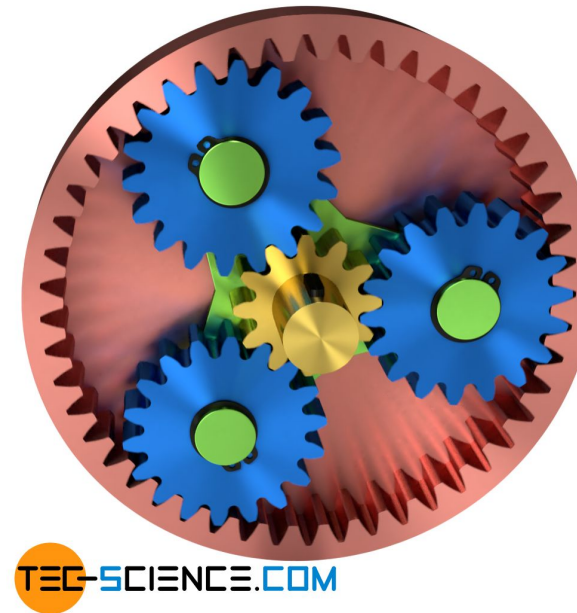
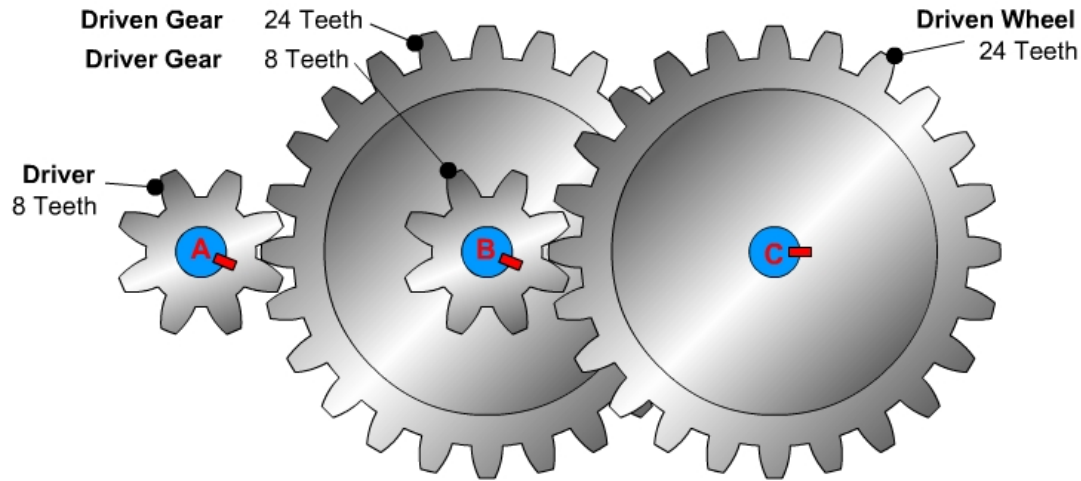
Quite ambiguous, Two main meanings:

- Backward-Efficiency
- Coulomb friction(or, reverse drive starting torque)
- Joint (reflected) inertia

Let's just use the word 'Back-drivability' as Friction, NOT inertia.

Select Reducer Type

- Why not serial, multiple spur gears?
 - Not only moment, but also radial load. Friction!
 - External gears are less efficient than internal gears.

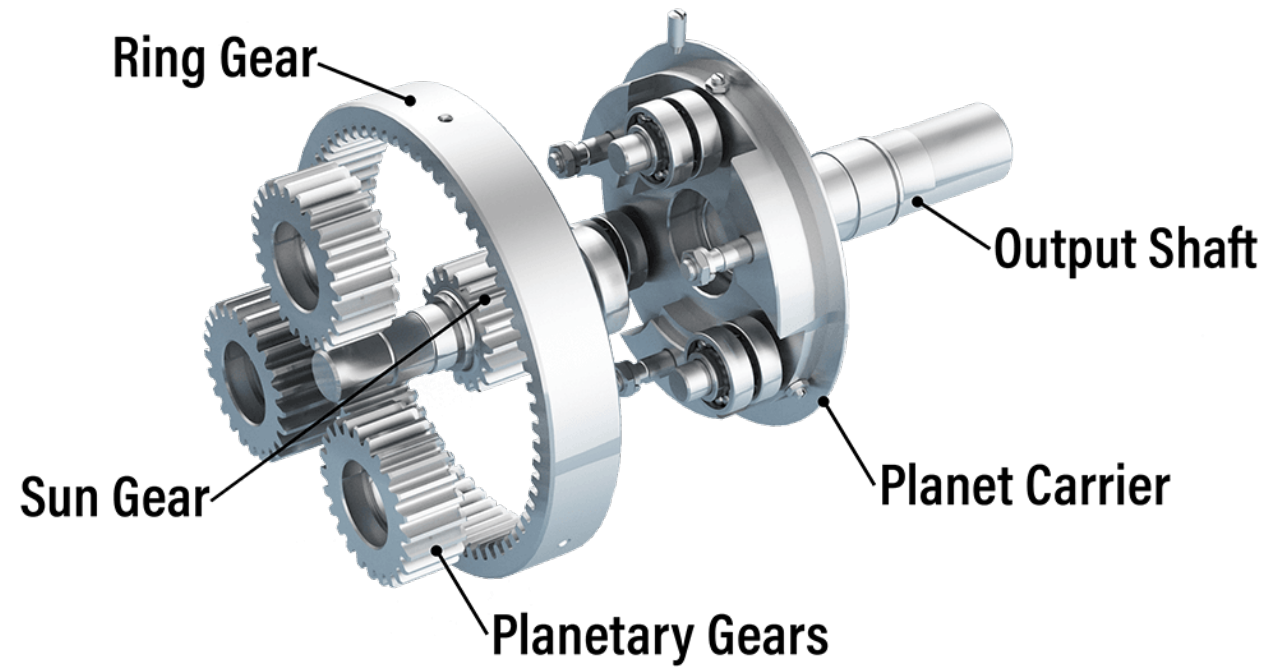
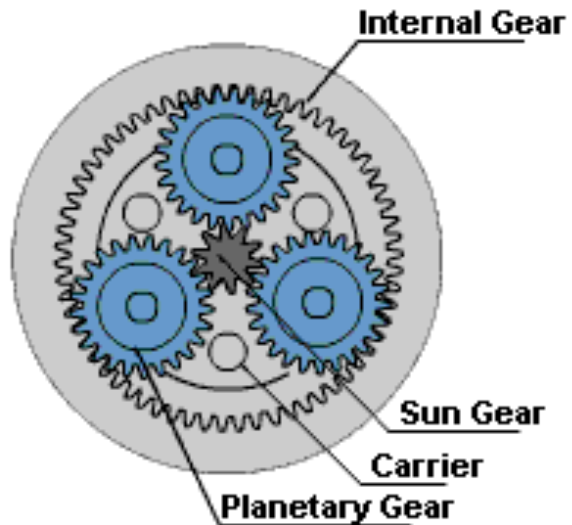


Select Reducer Type

- Single-stage Planetary Drive

Pros : High efficiency / Easy to design

Cons : Limited reduction ratio (<10) / Low torque density



Select Reducer Type

- 2-stage Planetary Drive

Pros : High efficiency

Cons : Limited reduction ratio / Many parts / Low torque density

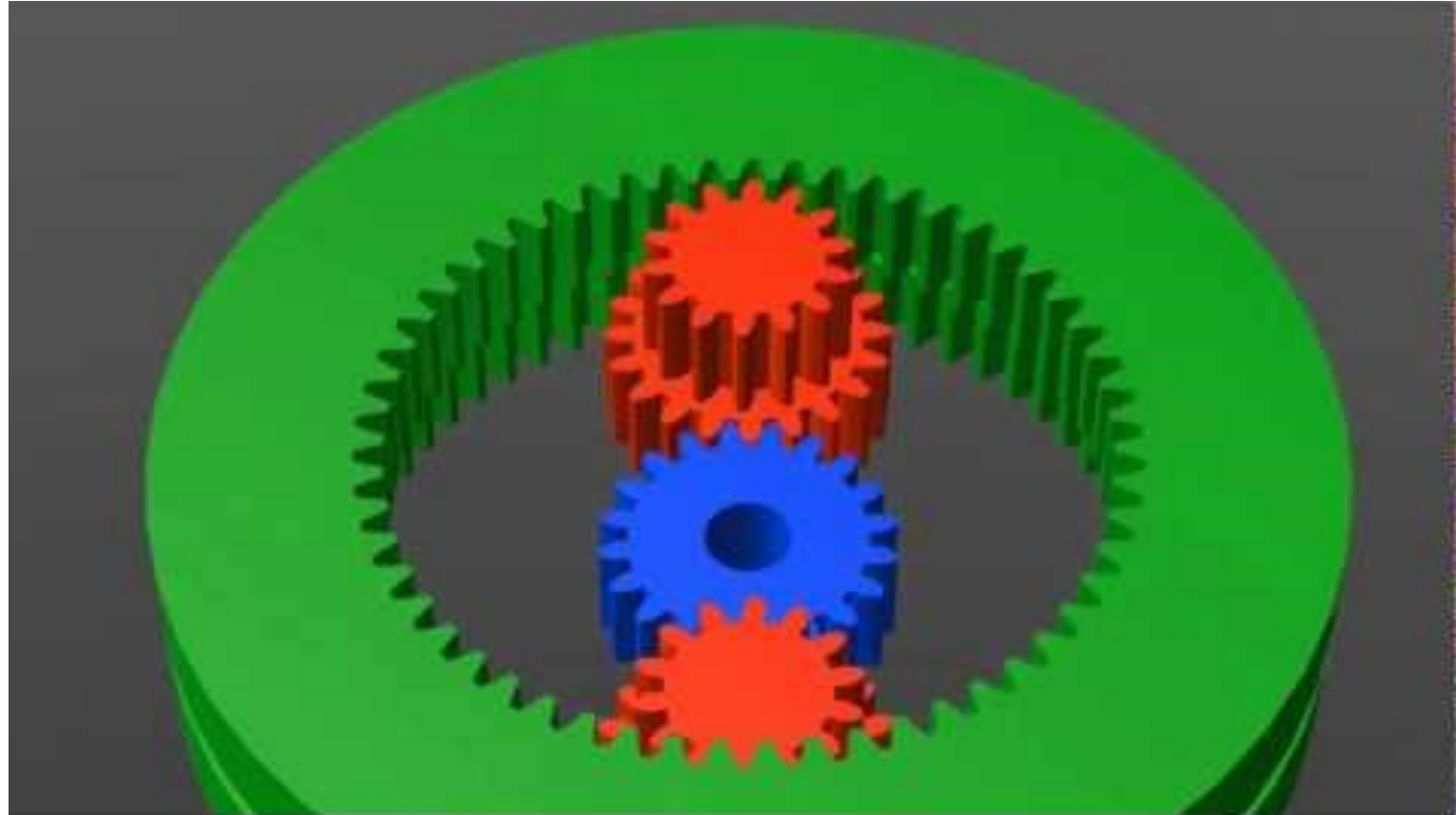


Select Reducer Type

- 3K Compound Planetary Drive

Pros : Wide range of reduction ratio / High torque density(compare to 2-stage PLD)

Cons : Low efficiency



Select Reducer Type

- Cycloidal Drive

Pros : High Efficiency(Ideal condition)

Cons : Needs VERY high precision for manufacturing



Select Reducer Type

- Harmonic Drive

Pros : High torque density

Cons : Low efficiency / VERY HARD to design & manufacture



To be Continue...